

32S3 — Relational Brackets, Completion Geometry, and $I \rightarrow P$ Admissibility

Professional reading edition of the evolving public MKUFT canon

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1. Purpose

The Module 32 family already distinguishes changing state-space address, reachable-state geometry, load-bearing relations, scale-separated freedom and capability, relational addressability, individuality parity, and temporal continuity.

This supplement sharpens a common mathematical object behind three questions:

1. **compatibility** — which local or distributed states can coexist under the declared relations;
2. **completion** — what could lawfully occupy a missing or unobserved address given the relations that remain;
3. **I→P admissibility** — whether a declared information/relation variable changes the support or weighting of later physical transitions after the strongest adequate physical baseline is controlled.

The shared object is a typed **constraint-to-admissibility grammar**, not a claim that every domain uses one physical mechanism.

2. Local coordinates, geometric overlap, and typed relation scopes

Let a declared system at time t have addressed components or regions

$$\mathcal{U}_t = \{U_1, \dots, U_n\},$$

with local state spaces

$$\mathcal{F}_t(U_i).$$

The ambient assignment space is

$$\mathcal{L}_t = \prod_{i=1}^n \mathcal{F}_t(U_i).$$

This product is bookkeeping for simultaneous coordinate assignments. It does **not** assert statistical independence, causal independence, separability, or physical isolation of the factors.

Where physical/geometric overlap is meaningful, let

$$N_t = N(\mathcal{U}_t)$$

be the nerve of the cover. The nerve is useful for genuinely local geometric relations, but it is not the whole bracket.

Define a broader typed relation-scope family

$$\mathfrak{E}_t = \{r_1, \dots, r_m\}.$$

Each relation r has a non-empty support

$$S(r) \subseteq \{1, \dots, n\},$$

plus a declared relation type, scale, role structure where direction/asymmetry matters, and measurement rule.

A relation scope may represent geometric overlap, regulatory coupling, system–environment coupling, temporal linkage, contextual compatibility, semantic/functional relation, or another operationally defined relation class. A non-geometric scope does **not** by itself imply a nonlocal physical mechanism or a new layer.

Geometric overlap relations form at most a typed subfamily

$$\mathfrak{E}_t^{\text{geom}} \subseteq \mathfrak{E}_t.$$

For a relation r and coordinate i in $S(r)$, let

$$\eta_{i \rightarrow r} : \mathcal{F}_t(U_i) \rightarrow \mathcal{Y}_{r,i}$$

extract or translate the state information relevant to that relation. For a genuine geometric overlap, η may reduce to an ordinary restriction map onto the shared domain. For other relation types it must be separately defined and justified.

Spatial overlap is one way for relations to be load-bearing; it is not assumed to be the only way.

3. Relational bracket and compatible assignment region

For assignment

$$a_t = (a_1, \dots, a_n) \in \mathcal{L}_t,$$

define a typed residual for each declared relation

$$e_r(a_t) = \Psi_r \left(\{ \eta_{i \rightarrow r}(a_i) \}_{i \in S(r)} \right) \geq 0,$$

where Ψ_r is defined so that $e_r=0$ means exact compatibility under that relation and larger values mean larger disagreement in the declared residual geometry.

Let ϵ_r be a preregistered tolerance in the same units as e_r . Define

$$\chi_B(a_t) = \mathbf{1}[e_r(a_t) \leq \epsilon_r \quad \forall r \in \mathfrak{E}_t].$$

The componentwise test is primary. Different residual types must not be summed merely because they are numerical.

The bracket selects the compatible assignment region

$$\mathfrak{E}_{B,t} = \{a \in \mathcal{L}_t : \chi_B(a) = 1\} \subseteq \mathcal{L}_t.$$

For reporting, retain

$$\mathbf{e}_B(a_t) = (e_r(a_t))_{r \in \mathfrak{E}_t}.$$

Only after each residual has a lawful scale $s_r > 0$ may a dimensionless aggregate be formed:

$$\tilde{e}_r = \frac{e_r}{s_r},$$

$$E_B(a_t) = \left[\frac{\sum_{r \in \mathfrak{E}_t} w_r \tilde{e}_r(a_t)^p}{\sum_{r \in \mathfrak{E}_t} w_r} \right]^{1/p}, \quad w_r \geq 0, \quad p \geq 1.$$

If no lawful normalisation exists, do not manufacture E_B ; retain the typed residual vector.

Define

$$\mathfrak{B}_t = (\mathcal{U}_t, N_t, \mathfrak{E}_t, \mathcal{F}_t, \eta_t, \Psi_t, \Theta_t),$$

where Θ_t contains declared tolerances, normalisations, comparison rules, and representation assumptions.

The bracket is not a container surrounding an object. It specifies **which coordinates participate in which relations and what joint assignments those relations permit**.

4. Completion geometry — a missing address as a conditional fiber

Suppose coordinate k is missing, unobserved, or intentionally ablated while the remaining partial assignment

$$a_{-k}$$

is retained.

For candidate completion

$$x \in \mathcal{F}_t(U_k),$$

define

$$a_{-k} \oplus_k x = (a_1, \dots, a_{k-1}, x, a_{k+1}, \dots, a_n).$$

Separate relations that touch the missing address from those that do not:

$$\mathfrak{E}_{k,t} = \{r \in \mathfrak{E}_t : k \in S(r)\},$$

$$\mathfrak{E}_{-k,t} = \{r \in \mathfrak{E}_t : k \notin S(r)\}.$$

Define the retained-background test

$$\chi_{B,-k}(a_{-k}) = \mathbf{1}[e_r(a_{-k}) \leq \varepsilon_r \quad \forall r \in \mathfrak{E}_{-k,t}]$$

for residuals well-defined without coordinate k , and the local completion test

$$\chi_{B,k}(a_{-k} \oplus_k x) = \mathbf{1}[e_r(a_{-k} \oplus_k x) \leq \varepsilon_r \quad \forall r \in \mathfrak{E}_{k,t}].$$

When

$$\chi_{B,-k}(a_{-k}) = 1,$$

define

$$\Omega_{k,t}^{\text{comp}}(a_{-k}) = \{x \in \mathcal{F}_t(U_k) : \chi_{B,k}(a_{-k} \oplus_k x) = 1\}.$$

If the retained background is inconsistent, a missing-address inference is confounded. If

$$\mathfrak{E}_{k,t} = \emptyset,$$

the bracket supplies no relation constraining that coordinate; it is unconstrained by this model rather than positively specified.

4.1 Fiber form

Let

$$\pi_{-k} : \mathcal{L}_t \rightarrow \prod_{i \neq k} \mathcal{F}_t(U_i)$$

forget coordinate k , and let

$$\pi_k : \mathcal{L}_t \rightarrow \mathcal{F}_t(U_k)$$

select it.

The compatibility fiber over the retained assignment is

$$\mathfrak{F}_{k,t}(a_{-k}) = \mathfrak{E}_{B,t} \cap \pi_{-k}^{-1}(a_{-k}),$$

and

$$\Omega_{k,t}^{\text{comp}}(a_{-k}) = \pi_k(\mathfrak{F}_{k,t}(a_{-k})).$$

This is standard fiber/projection mathematics. Its role here is operational: **hold the retained relational address fixed and ask what values remain admissible at the missing coordinate.**

4.2 Four distinct absence cases

1. **Narrow completion:** the completion set is non-empty and small under a lawful uncertainty/diameter measure.
2. **Broad completion:** the completion set is large; the address remains structured but substantially ambiguous.
3. **Local incompatibility:** the retained background is compatible but no tested value satisfies every relation touching the missing address.
4. **Background inconsistency:** relations not involving the missing address are already failing, so local completion cannot be interpreted cleanly.

Thus:

physically absent does not imply relationally unspecified; relationally specified does not imply physically occupied.

5. Ambiguity, persistence, and temporal mismatch exposure

Where a lawful measure μ_k exists,

$$A_{k,t}^{\text{comp}} = \log \left(1 + \frac{\mu_k(\Omega_{k,t}^{\text{comp}})}{\mu_{0,k}} \right)$$

is a completion-ambiguity index. It is not automatically information, energy, or cost.

For a time interval over which the bracket observables remain lawfully comparable,

$$P_B(t; \tau) = \frac{1}{\tau} \int_t^{t+\tau} \chi_B(a_s) ds$$

may be used as a persistence fraction, with a preregistered requirement such as

$$P_B(t; \tau) \geq 1 - \delta_B.$$

If relation scopes, state spaces, encodings, tolerances, or comparison rules change materially, use Module 31 transport/comparison discipline, partition the interval, or do not report one persistence scalar.

Where a lawful dimensionless aggregate $E_B(t)$ exists,

$$D_B(t; T) = \int_{t-T}^t E_B(s) ds$$

defines mismatch exposure. It has units of time when E_B is dimensionless and is not automatically energy, entropy, damage, disease burden, or maintenance cost.

Mismatch earns a **cost** interpretation only if it predicts or causally changes a separately measured typed cost object

$$\mathbf{c}_t = (c_t^{(1)}, \dots, c_t^{(m)})$$

under held-out or interventional testing.

6. Addressed constraint-to-admissibility operator

At declared layer/address λ , let

$$\mathfrak{X}_\lambda = \bigsqcup_{\alpha \in A_\lambda} \{\alpha\} \times \mathcal{X}_{\lambda,\alpha}$$

be the addressed state family.

Let

$$\mathbf{r}_{\lambda,t} \in \mathcal{Z}_{\lambda,t}^R$$

denote the realised typed relation-state object generated by the declared relation scopes and current organisation. $\mathcal{Z}^R$ is implementation-specific; it is not assumed to be one universal relation space.

For current construction address α_t , define schematically

$$\mathcal{Q}_{\lambda,t} : \mathcal{Z}_{\lambda,t}^R \times \mathcal{X}_{\lambda,\alpha_t} \times \mathcal{H}_{\lambda,t} \longrightarrow 2^{\mathfrak{X}_\lambda},$$

with

$$\mathcal{Q}_{\lambda,t}(\mathbf{r}, x, h) = \Omega_{\lambda,t}^{\text{adm}}.$$

The output is an **addressed admissible region** and may include transitions into a different construction/state-space address where the implementation permits that change.

This operator is a scaffold, not a universal physical law. It covers as typed specialisations compatible completion, ordinary reachable-state restriction, scale-conditioned loss of lower-level independent addressability, and candidate $I \rightarrow P$ changes in later physical transition support or weighting.

Bracket and reach are two views of constraint-shaped admissibility.

7. $I \rightarrow P$ as an admissibility question, not a message-pipe assumption

Let

$$p_t \in \mathcal{P}_t$$

be measured physical state, H_t^P the relevant measured physical history, and

$$i_t \in \mathcal{I}_t$$

a candidate information/relation variable.

Represent the empirical question by

$$K_P(dp_{t+\Delta} \mid p_t, H_t^P, i_t).$$

The corresponding support is

$$\Omega_{P,t}^{\text{adm}}(i_t) = \text{supp} K_P(\cdot \mid p_t, H_t^P, i_t).$$

For a preregistered lawful relation deformation

$$i_t \rightarrow \tilde{i}_t,$$

a generic distributional effect score is

$$\Lambda_{I \rightarrow P} = \mathbb{E}[D_P(K_P(\cdot \mid p, H, i), K_P(\cdot \mid p, H, \tilde{i}))].$$

A support-level test may instead use

$$d_\Omega(\Omega_P^{\text{adm}}(i), \Omega_P^{\text{adm}}(\tilde{i})) > \eta_\Omega.$$

A non-zero conditional difference is not automatically causal. If i cannot be changed or causally identified independently of relevant P variables, report predictive association rather than causal $I \rightarrow P$ evidence.

The existing physical-only null remains controlling:

$$H_0 : Y_{t+\Delta} \perp\!\!\!\perp I_t \mid P_t, H_t^P.$$

If

$$i_t = f(P_t, H_t^P)$$

under an adequate physical description, the I -layer variable may remain a useful relational abstraction or compression, but independent $I \rightarrow P$ dynamics have not been established.

A stronger independent $I \rightarrow P$ candidate requires representation robustness, a strongest adequate P baseline, held-out predictive or interventional gain, relation-specific deformation that does not merely introduce an uncontrolled P change, ordinary-limit recovery, and an explicit falsifier.

8. Operational semantic load-bearingness

The word semantic should not mean merely meaningful to an observer.

A declared relation earns **operational semantic load-bearingness** only when preserving versus disrupting that relation changes a preregistered system-relevant future observable, action distribution, viability condition, admissible state support, or other consequence while relevant lower-order statistics and carrier details are controlled as far as the implementation permits.

This is adjacent to established intervention-based semantic-information work. MKUFT does not claim invention of causal semantics.

If changing the relation does not change the tested future physical or functional possibilities, do not call that relation load-bearing for that claim.

9. Fiber–quotient scale handoff and realisation classes

The compatibility region supports complementary zoom-in and zoom-out questions.

9.1 Zoom in — completion fiber

Fix retained coordinates and ask which states remain admissible at a missing coordinate:

$$\mathfrak{F}_{k,t}(a_{-k}) = \mathfrak{C}_{B,t} \cap \pi_{-k}^{-1}(a_{-k}),$$

$$\Omega_{k,t}^{\text{comp}} = \pi_k(\mathfrak{F}_{k,t}).$$

9.2 Zoom out — quotient

Let a declared higher-scale readout be

$$R_{\ell \rightarrow L} : \text{Dom}(R_{\ell \rightarrow L}) \subseteq \mathfrak{C}_{B,\ell} \longrightarrow \mathcal{Y}_L.$$

Define

$$a \underset{R}{\sim} a' \iff R_{\ell \rightarrow L}(a) = R_{\ell \rightarrow L}(a'),$$

or a tolerance-based analogue. Then

$$\mathfrak{C}_{B,\ell} / \underset{R}{\sim}$$

groups compatible lower-scale assignments that are indistinguishable under the declared higher-scale readout.

The quotient is standard mathematics and is not automatically a physically autonomous object.

9.3 Macrostate realisation fiber

For

$$y \in \text{Im}(R_{\ell \rightarrow L}),$$

define the compatible realisation class

$$\mathcal{M}_y = \mathfrak{C}_{B,\ell} \cap R_{\ell \rightarrow L}^{-1}(y).$$

For any compatible a with $R(a)=y$,

$$\mathcal{M}_y = [a]_{\underset{R}{\sim}}.$$

Thus one higher-scale readout state can correspond to many lower-scale compatible realisations.

A lower-scale trajectory may therefore change while the higher-scale readout remains fixed:

$$a_t \neq a_{t'}, \text{ but } R(a_t) = R(a_{t'}) = y,$$

provided both states remain in $\mathcal{M}_y$.

This gives a precise systems meaning to **lower-level turnover with higher-level persistence**. It does not by itself prove metaphysical identity, personal identity, or an autonomous macro cause. The macrostate must still pass the Module 32S promotion tests, and any same-self claim remains governed by Modules 32S1 and 32S2.

When a trajectory leaves $\mathcal{M}_y$, the declared macrostate changes under this readout. Whether the individuality persists through that macrostate change is a separate identity-continuity question.

9.4 Promotion and recursive handoff

A higher-scale effective object is promoted only after persistence, interventional load-bearingness, new viable capability, predictive compression, and boundary/closure specificity are demonstrated.

Then schematically

$$\mathcal{L}_\ell \supseteq \mathfrak{C}_{B,\ell} \longrightarrow \mathfrak{C}_{B,\ell}/\sim_R \xrightarrow{\text{promotion tests}} O_L.$$

To use the promoted whole at the next scale, declare

$$\iota_{L \rightarrow L^+} : \mathcal{O}_L \rightarrow \mathcal{X}_{L^+}^{\text{const}},$$

with

$$\iota_{L \rightarrow L^+}(O_L) \in \mathcal{X}_{L^+}^{\text{const}}.$$

That typed constituent may then participate in the next bracket's relation scopes.

The recursive grammar is

```
local state product
→ relation-compatible subset
→ quotient under declared higher-scale readout
→ effective-object promotion when earned
→ typed constituent in the next relational bracket
```

while the complementary completion query is

```
relation-compatible subset
→ fix retained coordinates
→ take a projection fiber
→ recover admissible completion of the missing coordinate
```

Fiber and quotient are two mathematical views of the same compatibility geometry, not two physical mechanisms.

10. Relation to the S–I–P–O architecture

The bracket formalism can implement a **restricted operational mirror** of the S–I–P–O architecture without identifying the mathematical scaffold with the ontology.

At a declared implementation:

- **S-like role:** an addressed ambient possibility family supplies candidate states;
- **I-like role:** typed relations constrain which assignments/transitions are admissible;
- **P role:** a physical system realises one actual state or trajectory inside the physically admissible region;
- **O role:** an observer-positioned measurement/registration map returns bounded records or observables.

A higher-scale readout $R_{(\ell \rightarrow L)}$ used to define an endogenous macro variable is **not automatically the O-layer**. It may be a P-level effective description. It belongs to O only where the map represents observer-positioned registration under the definitions of the master spine.

This distinction matters because otherwise coarse-graining by an analyst could be mistaken for a real scale transition in the system.

The formalism therefore sharpens, but does not prove, the candidate I→P edge:

I-like relational organisation can be represented as changing the admissible physical state/transition region; independent I→P physics is claimed only if the relation contributes content beyond an adequate P-state/history account.

11. Holography as a deformation-tested physical comparator

Optical holography is useful because a spatial reconstruction can depend on distributed phase relations and a declared propagation/readout transform.

A simple scalar scaffold is

$$U_z = \mathcal{H}_z [A(x, y) e^{i\phi(x, y)}].$$

Under local mask $M(x, y)$,

$$U_z^{(M)} = \mathcal{H}_z[M(x, y)A(x, y)e^{i\phi(x, y)}].$$

Metasurface holography provides a literal example in which local element orientation encodes phase and the reconstructed spatial object appears only after the relevant optical transform.

The MKUFT use is methodological:

If another system is claimed to possess holographic or distributed reconstruction geometry, specify the encoding variables, transform/readout, and predicted response to local masking or phase deformation before calling the analogy physical.

Visual similarity, sacred geometry, or the phrase the whole is in every part is insufficient.

12. Empirical P-layer comparators

These examples remain P-layer comparators, not evidence for an independent I layer.

12.1 Boundary geometry and interior address

Guruciaga et al. (2026) report that boundary geometry controls polar-defect configuration in a model of the mouse epiblast and that predicted defect positions correspond to lumen nucleation sites. The narrow relevance is that **relations at a boundary can constrain the location of an interior physical event**.

12.2 Physical removal with persistent higher-order representation

Schone et al. (2025) report stable cortical hand/finger representations after arm amputation. The permitted structural statement is that **removal of a physical endpoint need not immediately erase every higher-order physical representation associated with that endpoint**. The surviving nervous system remains a physical implementation.

13. Discriminating tests

13.1 Relation-family and representation robustness

Change the lawful cover, coordinate chart, encoding, or higher-order relation representation while preserving the declared underlying object. Compare geometric-nerve, hypergraph/relation-scope, or other justified representations where appropriate. If the effect appears only under one arbitrary representation, reduce the claim.

13.2 Completion prediction

Remove, hide, or ablate a local occupant while preserving surrounding relations as far as possible. Verify retained-background compatibility, predict the completion fiber from relations whose support includes the missing coordinate, then test restored/repopulated outcomes against the preregistered set.

13.3 Surrounding-relation deformation

Preserve the candidate occupant state while changing selected surrounding or contextual relations. The completion fiber or later realised state should deform in the predicted direction if those relations are load-bearing.

13.4 Higher-order relation test

Compare pairwise-only models with preregistered higher-order relation scopes. Claimed higher-order compatibility must add held-out predictive/interventional value.

13.5 Realisation-class persistence test

For a proposed macrostate y , perturb lower-scale variables inside the predicted realisation class M_y . The macrostate should remain stable under within-class perturbations and change in the predicted way when the trajectory crosses the class boundary. Compare against alternative readouts and arbitrary coarse-grainings.

13.6 Temporal mismatch test

Compare instantaneous residuals against lawful cumulative mismatch exposure for held-out prediction of later closure failure or independently measured cost. If accumulated exposure adds no value, keep the simpler instantaneous model.

13.7 I→P relation-scramble test

Preserve as much carrier-specific and lower-order physical structure as practical while disrupting the proposed relation. Test the later transition kernel/support against the adequate P-only model. If causal isolation cannot be achieved, do not call the result causal I→P evidence.

13.8 Holographic deformation comparator

Predeclare the analogue of encoding, phase/amplitude deformation, mask, propagation/readout, and predicted reconstruction degradation. Compare with simpler distributed-network alternatives.

13.9 Scale-promotion test

Compare the proposed quotient/readout with alternative coarse-grainings. Promote a higher-scale object only if the full Module 32S effective-unit test passes.

14. Failure and reduction conditions

Reduce this refinement if:

- local or pairwise variables predict the outcome as well as the declared relation-scope architecture;
- completion fibers fail held-out occupancy, regeneration, recovery, or state-resolution tests;
- unrelated background inconsistency is mistaken for failure of a missing-address completion;
- the result depends on an arbitrary cover, relation representation, coordinate system, or encoding and fails representation-preserving translation;
- higher-order scopes add no value beyond pairwise relations;
- proposed macro realisation classes fail perturbational stability or an alternative coarse-graining predicts equally well with less structure;
- temporal mismatch is accumulated across non-comparable addresses without lawful transport or adds no value beyond instantaneous state/history;
- relation scrambling produces no predicted consequence;
- a claimed causal $I \rightarrow P$ result cannot separate relation change from uncontrolled P-layer change;
- the I variable reduces to the adequate P-state/history description where independent $I \rightarrow P$ dynamics were claimed;
- the quotient/readout is observer-only compression with no endogenous interventional or predictive gain;
- a holographic claim fails the preregistered transform/deformation response or is explained equally well by a simpler distributed model;
- cross-scale recurrence survives only as naming or visual resemblance rather than typed relation and deformation equivalence.

If these failures occur, use the simpler local, physical, pairwise, network, coarse-grained, or ordinary dynamical description.

15. Prior-art and novelty boundary

The broad ingredients used here are established and are not claimed as MKUFT inventions.

Relevant neighbours include:

- nerve constructions, sheaves, restriction/gluing, fibers, projections, quotients, equivalence classes, coarse-graining, and multiple realisability;
- hypergraphs and higher-order networks for arbitrary non-pairwise relation scopes;
- sheaf-theoretic local-to-global analysis over contextual measurement covers, including non-locality/contextuality work;
- intervention-based semantic information;
- causal emergence and information conversion across scales;
- optical holography and phase encoding/reconstruction;
- constraint-closure, morphogenetic, neural, and systems-biology methods.

The candidate MKUFT contribution is narrower:

integrate addressed changing-state-space discipline, typed relation scopes, completion fibers, macrostate realisation classes, quotient-based scale readout with separate promotion tests, persistent mismatch exposure, and the $I \rightarrow P$ physical-null boundary into one operational audit in which relationally specified absence, lower-level turnover with higher-level persistence, and changed physical admissibility are tested inside one constraint-to-admissibility grammar.

Historical priority for that exact integration is not asserted without broader literature review. Module 32A owns the evolving novelty audit.

16. References

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17. Compressed rules

Geometric overlap is one typed relation class, not the whole relation architecture.

The ambient product of local state spaces is a coordinate scaffold, not an independence assumption.

The bracket is the typed relation-scope architecture that selects compatible assignments.

A missing-address question is a fiber of the compatible region; a higher-scale readout is a quotient of it.

A macrostate is realised by the compatible preimage of its readout value; lower-level states can change inside that realisation class while the macrostate remains fixed.

Physically absent does not imply relationally unspecified; relationally specified does not imply physically occupied.

Persistent mismatch is not automatically cost. Measure mismatch and cost separately, then test their coupling.

The $I \rightarrow P$ question can be posed as changed physical transition support or weighting without assuming a hidden force, carrier, or message pipe.

A non-zero relational effect may remain fully P -realised; a non-zero conditional difference is not automatically causal.

Readout does not create a scale transition. Promotion requires persistence, intervention, new capability, predictive compression, and boundary specificity.

Holography is a deformation-tested physical comparator, not an ontological shortcut.

At every justified scale: typed relations constrain addressed admissibility; stable compatible structure can be promoted to an effective object; that object may become one typed participant in the next bracket.